

Investment Strategy Challenge 2026 · Proposal to the NKE Private Wealth Investment Committee

The Asia Digital Bridge Legacy Note

Own Asia's AI supply chain, from wafer to app, while every dollar of principal and every banked gain is kept for the next generation

USD 100Mn · 7-year USD note · 100% principal protected · 125% uncapped participation · Trade date 17 Sep 2026

Team [name] · Natixis Global Markets Institute

Agenda

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Executive summary

A 7-year USD note that turns high USD rates into protected, uncapped exposure to Asia's AI stack

100%

Principal back in 2033

Contractual floor at maturity (Natixis credit). A zero-coupon bond costing 67.6% funds it.

125%

Uncapped participation

In an equal-weight basket: Nikkei 225, TAIEX, KOSPI 200 and Hang Seng TECH, quanto USD

115 → 150%

Legacy Lock-in

Gains of +30% / +60% / +100% on any anniversary lock the floor at 115% / 130% / 150%

Recommendation. NKE invests USD 100Mn (5% of family wealth) in a Natixis note covering the four markets that make up Asia's AI supply chain: equipment, foundry, memory and platforms. The family's operating businesses already own the physical layer (data centres, power, real estate). The note adds the digital layer they lack, in USD, without picking a single winner. **In our 100,000-path simulation the note's median redemption is 130% and the worst 5% of outcomes still return 100%. Holding the basket directly would return 53% in that worst 5%, with a 29% chance of losing money.**

What it costs: no dividends, and the forgone 7Y bond yield. In 29% of simulated paths the family gets exactly its principal back. Mark-to-market can fall 17% in a joint equity-vol-rate shock before maturity.

The client: a third-generation family moving from operating to investing

NKE Private Wealth, the single-family office of the Chak family



Client profile

- Chak family, third generation; operating succession just completed
- Businesses: technology infrastructure, renewable energy plants and real estate across Japan and Greater China
- Family net worth > USD 2Bn
- **Mandate: USD 100Mn (about 5% of wealth) for one structured allocation**

Objectives



Capital appreciation for future generations

Long-horizon growth, not current income



Meaningful exposure to Asia's digital transformation

Enough weight to matter, not a token allocation



Bridge traditional Asian markets and new technology

Where the family has operating expertise



Downside managed for generational preservation

No permanent loss of the capital being passed on

Risk tolerance: Moderate

Can live with mark-to-market swings; cannot accept permanent capital loss

Key assumptions

1

Hold to maturity: a 7-year horizon, one succession cycle

2

Reports in USD; operating cash flows are in JPY, CNY and HKD

3

Professional investor; can hold OTC notes and accepts Natixis credit risk

4

No income need: the businesses provide cash flow, so no coupon

The family owns Asia's physical layer; the note adds the digital layer

Diagnosis: avoid adding to risks the family already carries

The AI value chain in Asia

Applications & platforms

AI models, cloud, e-commerce, super-apps

Hang Seng TECH

Semiconductors

Equipment · foundry · memory / HBM

Nikkei 225 · TAIEX · KOSPI 200

Physical infrastructure

Data centres · renewable power · real estate

*Already owned via family
businesses*

◀ *the note adds these two layers*

Keep principal
→ 100% protection

Add digital, not physical
→ AI-stack basket

No new FX risk
→ USD quanto

Don't pick one winner
→ 4-market basket

Bank gains for heirs
→ Legacy Lock-in

What the family's balance sheet already carries

- JPY, CNY and HKD currency exposure
- Greater China property cycle
- Power prices and interest-rate sensitivity in renewables
- Illiquid, concentrated, hard-asset wealth

Implication: no China property, utilities or REITs in the basket; pay out in USD; keep capital intact.

Asia builds the AI economy, but the 2026 rally makes entry risky

Market view as of the 17 Sep 2026 trade date



AI capex runs through Asia

> USD 600bn

Big-5 hyperscaler capex 2026E, mostly AI servers, GPUs and data centres

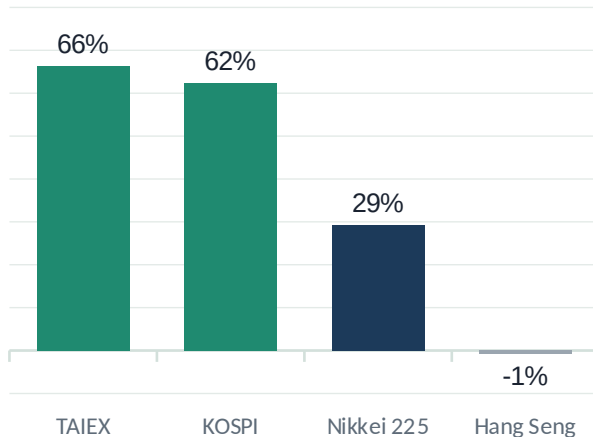
47%

Chips' share of Korean exports in 1–10 Sep 2026 (+270% YoY)

4 markets

Japan (equipment), Taiwan (foundry), Korea (memory), HK/China (platforms)

2026 YTD return (%), to late Sep



Implication: buying the momentum markets outright after +60% puts legacy capital at risk of a 2000-style reversal. Hong Kong tech has lagged, which gives a lower entry point.



Oil shock keeps USD rates high

- Fed hiked to 3.75–4.00% on 16 Sep 2026, its first hike in three years; the dots signal more
- Brent near USD 109 on the Iran war, with volatility elevated across Asia
- **USD 7Y Natixis funding of 5.75%: 67.6 cents today buys USD 1 in 2033**

Therefore: own the AI stack through a floor, in USD, for 7 years. High rates pay for the protection.

The Asia Digital Bridge Legacy Note

One note, four markets and the full AI stack, with a floor that only ratchets up

A 7-year, USD-denominated Natixis note. It returns 100% of principal at maturity plus 125% of the basket's gain, uncapped. Each anniversary, strong gains are locked in as a higher minimum redemption that can never fall back.



100% principal protection

Funded by a 7Y Natixis zero-coupon bond



125% participation, no cap

Full upside for the next generation



Legacy Lock-in ladder

+30% / +60% / +100% → floor 115 / 130 / 150%



Quanto USD

Local-market returns, no JPY/TWD/KRW/HKD risk

Index	Bloomberg	Wt	AI-stack role / bridge
Nikkei 225	NKY Index	25%	Traditional Japan Inc. + chip equipment (Tokyo Electron, Advantest)
TAIEX	TWSE Index	25%	Foundry and AI servers (TSMC, Hon Hai, MediaTek)
KOSPI 200	KOSPI2 Index	25%	Memory / HBM (SK hynix, Samsung Electronics)
Hang Seng TECH	HSTECH Index	25%	Platforms and AI apps (Tencent, Alibaba, Xiaomi)

Why these four: three traditional national benchmarks now driven by tech, plus one pure new-economy index. That is the bridge the family asked for. All four have deep listed futures and options, so Natixis can hedge at low cost. Equal weight keeps any one geopolitical risk (Taiwan, China regulation) to 25%.

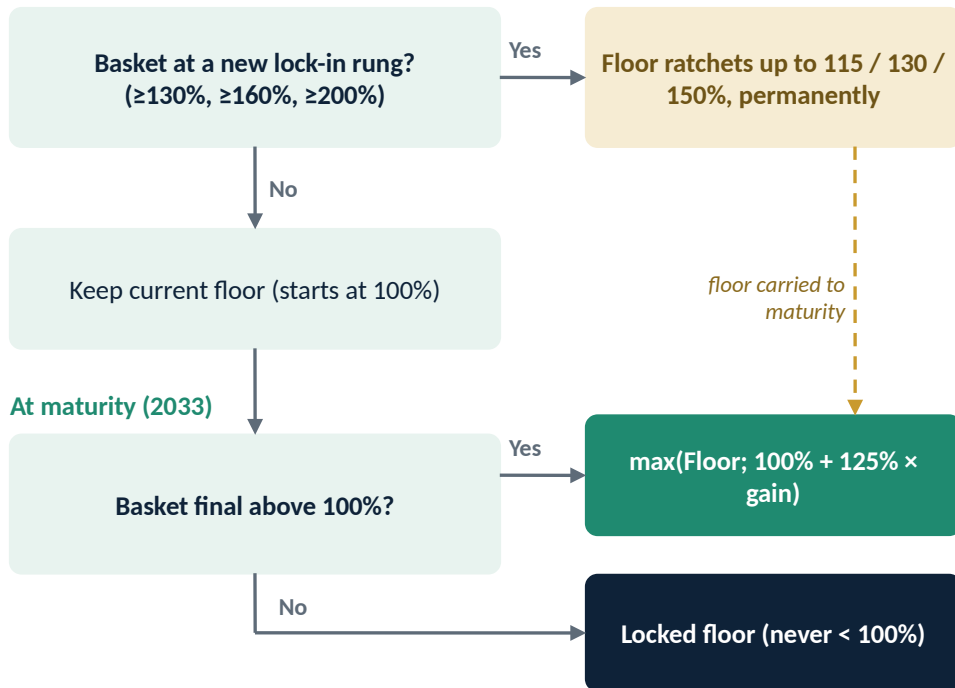
How the note works

Key terms and redemption logic

Key terms

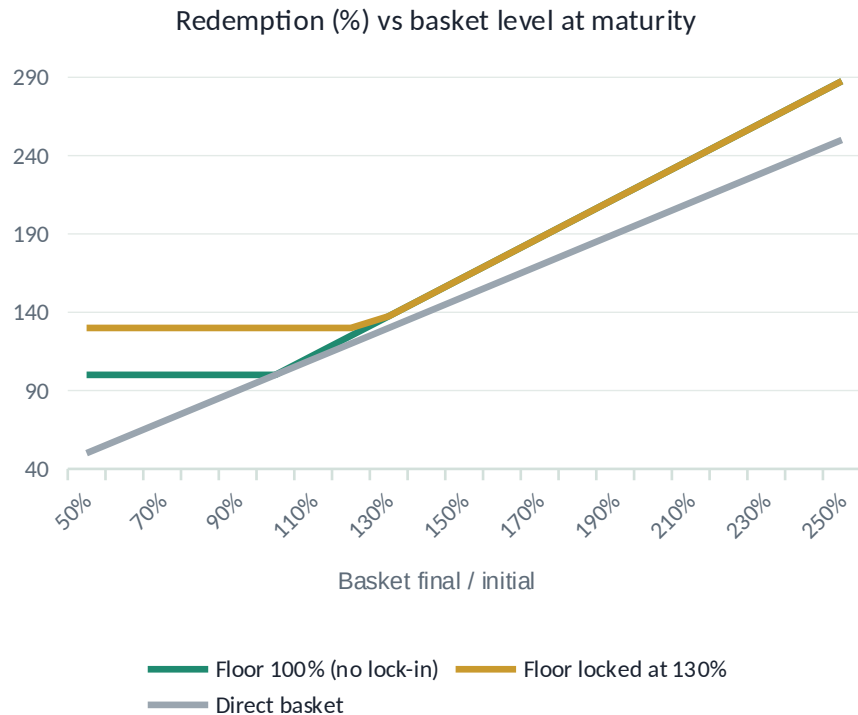
Issuer	Natixis S.A. (EMTN programme, private placement)
Notional / price	USD 100,000,000 at 100%
Trade / strike date	17 Sep 2026
Maturity	17 Sep 2033 (7 years)
Underlying	Equal-weight basket: NKY, TWSE, KOSPI2, HSTECH; quanto USD
Participation	125% of basket gain, uncapped
Lock-in dates	Annually, 17 Sep 2027 to 2032 (6 dates)
Lock-in ladder	Basket $\geq 130\%$ $\rightarrow$ floor 115%; $\geq 160\%$ $\rightarrow$ 130%; $\geq 200\%$ $\rightarrow$ 150%
Redemption	$\max[\text{Locked floor}; 100\% + 125\% \times \max(0, \text{Basket_final} - 100\%)]$
Liquidity	Natixis indicative secondary bid, daily

Each anniversary, years 1–6



Payoff and scenario analysis

Upside is uncapped; every outcome ends at or above 100%, and a boom that later busts still banks gains



Scenario (7 years)	Basket end	Peak	Note	Direct*
Steady AI build-out (+7% p.a. each)	161%	150%	176%	182%
AI boom then bust (2000-style)	92%	168%	130%	104%
Lost decade (Japan 1990s-style)	90%	100%	100%	102%
Early crash, slow recovery (2008-style)	130%	122%	138%	147%
China tech decoupling	125%	120%	131%	141%
Taiwan Strait shock in year 2	88%	105%	100%	100%

Read-across: the note gives up some upside in steady growth (no dividends) in exchange for a 130% exit in a boom-and-bust and 100% in a crisis. That is the trade-off a legacy portfolio should make.

*Direct = equal-weight basket bought outright, including ~1.8% p.a. dividends. Scenarios are stylised annual paths ([pricing/scenario2/analysis.py](#)).

Pricing: high USD rates fund an option budget of about 30%

Where the USD 100 goes, and the dials the family can turn

Use of USD 100 (% of notional)



Note

- Zero-coupon bond (100% floor): 67.6%
- Uncapped basket call (125%): 26.7%
- Legacy Lock-in ladder: 1.8%
- Natixis fee (35 bp p.a.): 2.5%
- Model reserve (skew, fwd vol): 1.5%

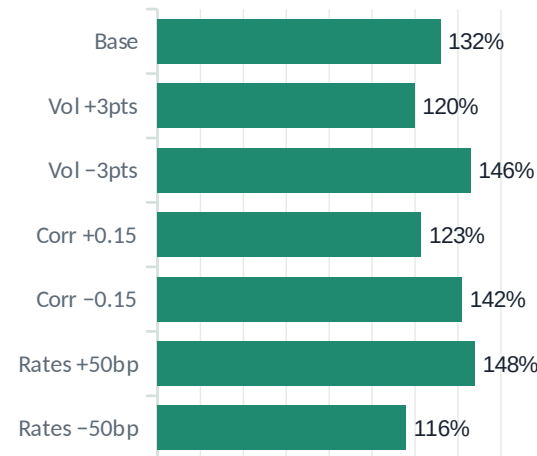
Structuring dials (fair participation)

Tenor	Funding	Budget	Particip.
3Y	5.52%	13.8%	86%
5Y	5.69%	22.4%	111%
7Y	5.75%	29.9%	132%
10Y	5.82%	39.7%	160%

Protection	Budget	Particip.
100%	29.9%	132%
95%	33.3%	146%
90%	36.7%	159%

Why 7Y and 100%: 5Y buys only 111%; 10Y locks capital beyond one succession cycle. A 90% floor adds 27% participation but breaks the preservation mandate.

Fair participation under shocks

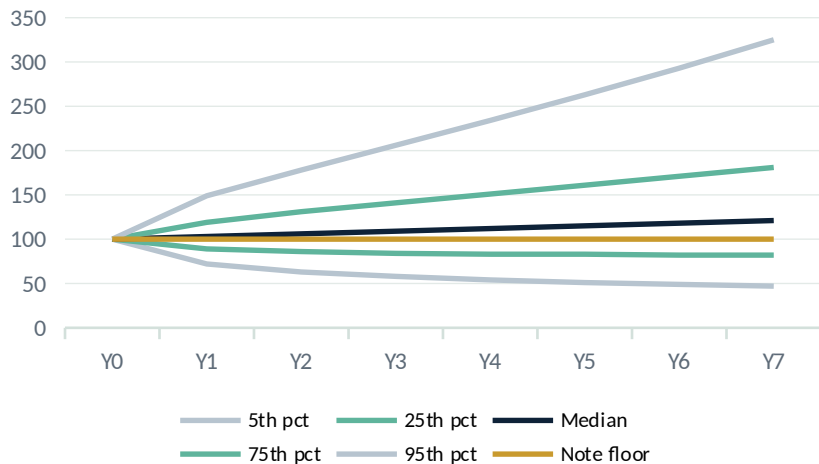


Design choices tested: no lock-in 140%; best-of rainbow 40/30/20/10 only 86%. We quote 125% vs 132% fair; the gap is a model reserve.

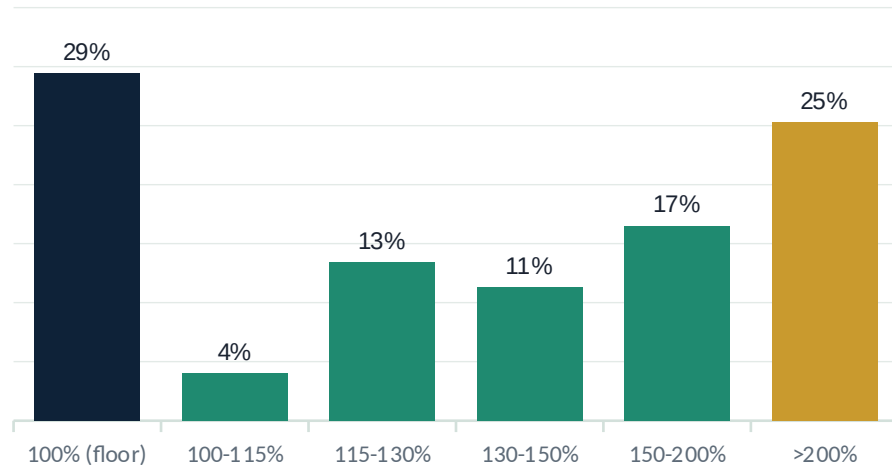
Back-testing: 100,000 fat-tailed paths over 7 years

Same median as owning the basket directly, without the left tail

Basket level fan chart (% of initial)



Distribution of note redemption at maturity (% of paths)



130%

Median redemption

6.5%

Mean IRR p.a.

59%

Paths locking $\geq 115\%$

100% vs 53%

5th pct: note vs direct

13%

Paths where lock-in beat final

Risk analysis: what can go wrong, and what we do about it

Investor risks, risk metrics and issuer hedging

Risk	What it means for the family	Mitigant
Issuer credit	Natixis default would remove the floor	Size capped by credit budget; track Natixis 5Y CDS
Opportunity cost	29% of paths return only principal	Lock-in banks gains; bonds elsewhere carry income
Mark-to-market	Value can dip before 2033; secondary bid/offer	Hold-to-maturity mandate; daily Natixis bid
Geopolitics	Taiwan Strait, China tech regulation	25% cap per market; floor absorbs the tail
Model / pricing	Vol, correlation, skew on the lock-in	1.45% reserve; sensitivities on p.7
Dividends / FX	No dividends; no local-FX upside	Priced into 125%; USD matches reporting

Greeks at issue (per USD 100 notional): delta USD 0.71 per 1% basket move • vega USD 0.88 per vol pt • DV01 USD -0.065 per bp. MTM risk before maturity is mostly rates: the 7Y bond floor has a duration of about 6.6.

Risk metrics (Unit 15)

Sizing: credit budget 3% of USD 2Bn = USD 60Mn ÷ 60% loss if Natixis defaults (40% recovery) = USD 100Mn max

10-day 99% VaR (MTM): 7.9%; ES 9.1%

Joint stress: equities -30%, vol +10pts, USD rates +100bp: -16.7% MTM. With rates -50bp: -8.4%

At maturity: loss is 0% by contract (ex-credit)

How Natixis hedges

- Delta: OSE/CME Nikkei, TAIFEX, KRX KOSPI 200 and HKEX HSTECH futures
- Vega: long-dated listed options and variance swaps
- Quanto: dynamic FX forwards and NDFs (TWD, KRW)
- Correlation / forward skew: kept as residual, covered by the reserve

Recommendation: invest USD 100Mn in the Legacy Note

For the Chak family

Objectives met: Growth (uncapped 125%), Asia digital exposure, a traditional-to-tech bridge, 100% floor

Complements, not duplicates: Adds the chips-and-platforms layer and USD; avoids property, power and local FX

Gains banked for heirs: The Legacy Lock-in turns a boom that later busts into a 130% exit

For Natixis

Hedgeable with listed instruments: Four of Asia's most liquid futures and options markets

Efficient funding: 7Y USD funding at the grid rate; 35 bp p.a. fee on a USD 100Mn ticket

Relationship: A long-term partnership with the next generation of a USD 2Bn family

Exit plan: all four exits defined



Profit target

Lock-in runs bank gains automatically. The investment committee reviews selling back if $MTM \geq 150\%$



Stop-loss

Not needed for market risk (contractual floor). Trigger on credit: Natixis 5Y CDS > 250bp; gap risk acknowledged



Thesis invalidation

Hyperscaler capex guided down for two straight quarters, or a Taiwan Strait blockade: review the position



Time stop

Maturity 17 Sep 2033, aligned with the family's next succession review

Appendix A1: market inputs and assumptions

Placeholders calibrated to typical levels; every one is a Bloomberg field to refresh at the trade date

Index	BBG ticker	Implied vol	Local rate	Div. yield	Quanto adj.	Q drift	Real-world drift
Nikkei 225	NKY Index	23%	3.23% (JPY grid)	1.7%	+0.5%	2.0%	5.0%
TAIEX	TWSE Index	26%	1.75%	2.6%	-0.2%	-1.1%	5.5%
KOSPI 200	KOSPI2 Index	29%	2.75%	1.9%	-0.3%	0.6%	5.0%
Hang Seng TECH	HSTECH Index	33%	5.00% (HKD)	0.9%	0.0%	4.1%	6.0%

Correlation matrix

	NKY	TWSE	KOSPI2	HSTECH
NKY	1.00	0.55	0.60	0.40
TWSE	0.55	1.00	0.65	0.45
KOSPI2	0.60	0.65	1.00	0.50
HSTECH	0.40	0.45	0.50	1.00

Funding: USD grid from the rules (7Y 5.75%); tenors interpolated linearly. Derivatives discounted at OIS $\approx$ funding - 45bp.

Fees: 35 bp p.a. (2.45% upfront equivalent).

Bloomberg to pull: OVDV (vol surfaces), CORR / HS (1Y and 3Y correlation), BDVD (dividends), SWPM (local curves), MEMB (constituents), CDSW (Natixis CDS).

Appendix A2: pricing and back-test methodology

Reproducible: every number in this deck comes from pricing/scenario2/analysis.py

1

Budget: Option budget = 100% – ZCB(7Y, USD funding) – fees. ZCB = $1 / 1.0575^7 = 67.6\%$.

2

Price: Risk-neutral quanto GBM, 4 correlated indices, annual steps (lock-in dates), 200,000 paths, common random numbers across bumps.

3

Solve: Bisection on participation so the PV of the payoff above 100% equals the budget. Quote below fair to hold a model reserve.

4

Simulate: Real-world monthly multivariate-t (df = 5) paths for fat tails and joint crashes, 100,000 paths, equity risk premium drifts.

5

Stress: Six stylised 7Y paths (boom-bust, lost decade, 2008-style, China tech decoupling, Taiwan shock, steady build-out).

6

Risk: Bump-and-revalue Greeks; delta-normal 10-day VaR/ES; full revaluation joint stress (spot, vol and rates together).

Historical back-test on Bloomberg data: export month-end PX_LAST for NKY, TWSE, KOSPI2, HSTECH (HSI spliced before 2015) and run backtest_bloomberg.py. It produces (a) a note struck every month with a full 7Y history and (b) a 12-month block bootstrap that keeps the cross-market correlation and crisis clustering that i.i.d. resampling destroys.

Appendix A3: alternatives we considered and rejected

Each fails at least one of the family's constraints

Alternative	Appeal	Why not for the Chak family
Worst-of autocallable on the same 4 indices	High fixed coupon	Barrier exposes principal to the single worst market (Taiwan or China tail); the family needs growth, not income
Best-of rainbow (40/30/20/10 by rank)	Weights the winning market	Participation falls to 86%: paying for dispersion the family does not need
10-year tenor	160% participation	Locks capital beyond one succession cycle; more credit exposure to one issuer
90% protection	159% participation	A 10% loss of generational capital is outside the mandate
Digital-asset (Bitcoin) sleeve	Direct crypto-tech exposure	50%+ vol would consume the option budget; reputational and custody questions for a legacy portfolio
Asia data-centre REITs / utilities	Close to family expertise	Duplicates the physical layer the family already owns through its businesses

Appendix A4: sources

- Natixis GMI, Investment Strategy Challenge 2026: game rules and USD funding grid
- [Goldman Sachs: Global AI investment forecast to exceed USD 1 trillion in 2026](#)
- [CreditSights: Hyperscaler capex 2026 estimates](#)
- [TechTimes: South Korea chips hit 47% of exports \(11 Sep 2026\)](#)
- [Seoul Economic Daily: KOSPI cedes global lead to Taiwan \(1 Oct 2026\)](#)
- [Advisor Perspectives: World markets watchlist \(21 Sep 2026\)](#)
- [CNBC: Fed rate decision, September 2026](#)
- [Allianz Research: AI capex cycle, war-proof for now \(Mar 2026\)](#)
- [Hang Seng Indexes: Hang Seng TECH Index factsheet](#)
- CUHK FINA4370A Unit 4 (volatility surface) and Unit 15 (trading proposal framework)